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Advancements in Relative Permeability Modifier Technologies for Sustainable Hydrocarbon Extraction

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Abstract

Relative Permeability Modifiers (RPMs) have proven to be a crucial technology in optimizing hydrocarbon recovery from reservoirs while minimizing water production. RPMs work by altering the wettability of rock surfaces within the formation, thereby modifying the flow characteristics of fluids through the porous media. The objective of this paper is to introduce a new class of RPMs that promises significant advancements over previous commercial solutions, for use in both sandstone and carbonate reservoirs.

To assess the suitability of RPM candidates for water conformance applications, comprehensive laboratory evaluations were conducted to assess their impact on fluid flow properties through porous media of different lithologies. This includes XRD analysis of core samples and conducting core flooding tests to measure changes in relative permeability as a function of pH, superficial velocity, and brine salinity. By comparing pre- and post-treatment results, engineers can gauge the effectiveness of RPM in modifying fluid behavior within the tested samples. In addition, compatibility tests to determine fluid-fluid interactions, and contact angle measurements to examine changes in fluid-solid interactions. Through meticulous lab-scale execution of tests, it was possible to gain valuable insights necessary to make better informed decisions prior to embarking on actual field deployments aimed at mitigating excessive water production issues.

The efficacy of these next generation RPM formulations was tested against a diverse set of sandstone and carbonate cores exhibiting air permeabilities spanning two orders of magnitude (0.5-1000 md). These experiments revealed an impressive ability to restrict water influx under varying temperature regimes (up to 275°F). Specifically, lab analysis indicated that the water permeability was effectively reduced in a range between 83 to 93%, albeit with notable sensitivity to certain environmental parameters: formation waters with dissolved acidic gases (pH < 5), hypersaline brines (>200,000 mg/L total dissolved solids), and high superficial velocities. Encouragingly, subsequent measurements confirmed negligible impairment to hydrocarbon transport properties following treatment, yielding average regain ratios exceeding 95% for gas and 70% for oil phases. The results provide a good basis for guiding technology deployment and with setting expectations or criteria for treatment success.

The series of laboratory tests performed validated the value proposition of these new generation RPMs and provides the confidence needed to proceed to field deployments. These breakthroughs collectively contribute to a sustainable tomorrow, reducing congenital water treatment at surface, optimizing reservoir performance, increasing operational efficiency, and reducing environmental impact — a multi-dimensional innovative solution critical for sustaining profitability across today's energy landscape.

Introduction

Excessive water production encountered in mature oil and gas fields generally causes economic and operational problems. This includes: Reduction in oil or gas production, extra costs for oil/water separation and handling, and disposal of large amounts of produced water (1). Relative permeability modifiers (RPMs), known to selectively reduce the permeability to water with minimum impact on the permeability to oil or gas, are a proven technology for resolving excessive water production from reservoirs with segregated oil and water flow (2).

Relative Permeability Modifier fundamentals

The basic criterion of an optimal RPM system reported in literature is that it should have preferably no permeability reduction for oil and significant permeability reduction for water. According to Dalrymple et al. (2000) an effective RPM is one with a residual resistance factor to water (RRFw) of 5 to 10 while maintaining a residual resistance factor to oil (RRFo) of 2 or less (see Figure 1–2).

$$RRF_w = \frac{K_{w\_before\ treatment}}{K_{w\_after\ treatment}} = \frac{\Delta P_{w\_after\ treatment}}{\Delta P_{w\_before\ treatment}} \gg 5,$$

$$RRF_o = \frac{K_{o\_before\ treatment}}{K_{o\_after\ treatment}} = \frac{\Delta P_{o\_after\ treatment}}{\Delta P_{o\_before\ treatment}} \leq 2,$$

where

Kw – effective water permeability, md Ko – effective oil permeability, md

ΔPw – average differential pressure for water flow @ SOR, psi

ΔPo – average differential pressure for oil flow @ SWI, psi

One criteria for acceptable RPM performance given by other researchers is the Normalized Fluid Resistance Ratio (NFRR):

$$NFRR = \frac{RRF_w}{RRF_o} \gg 2.5,\ provided\ that\ RRF_o \leq 2$$

Another criteria for acceptable RPM performance given by Stavland et al. (2000) is:

$$Optimum\ RPM = \frac{1}{RRF_o} - \frac{1}{RRF_w} \gg 0.3$$

Under normal circumstances ranges from 0 to 1, where 1 gives the most optimum RPM effect.

RRFw	1.0	2.0	3.0	4.0	5.0	6.0	8.0	10.0	12.0	14.0	16.0	18.0	20.0
% Water Regain	100.0%	50.0%	33.3%	25.0%	20.0%	16.7%	12.5%	10.0%	8.3%	7.1%	6.3%	5.6%	5.0%
% Water Reduction	0.0%	50.0%	66.7%	75.0%	80.0%	83.3%	87.5%	90.0%	91.7%	92.9%	93.8%	94.4%	95.0%

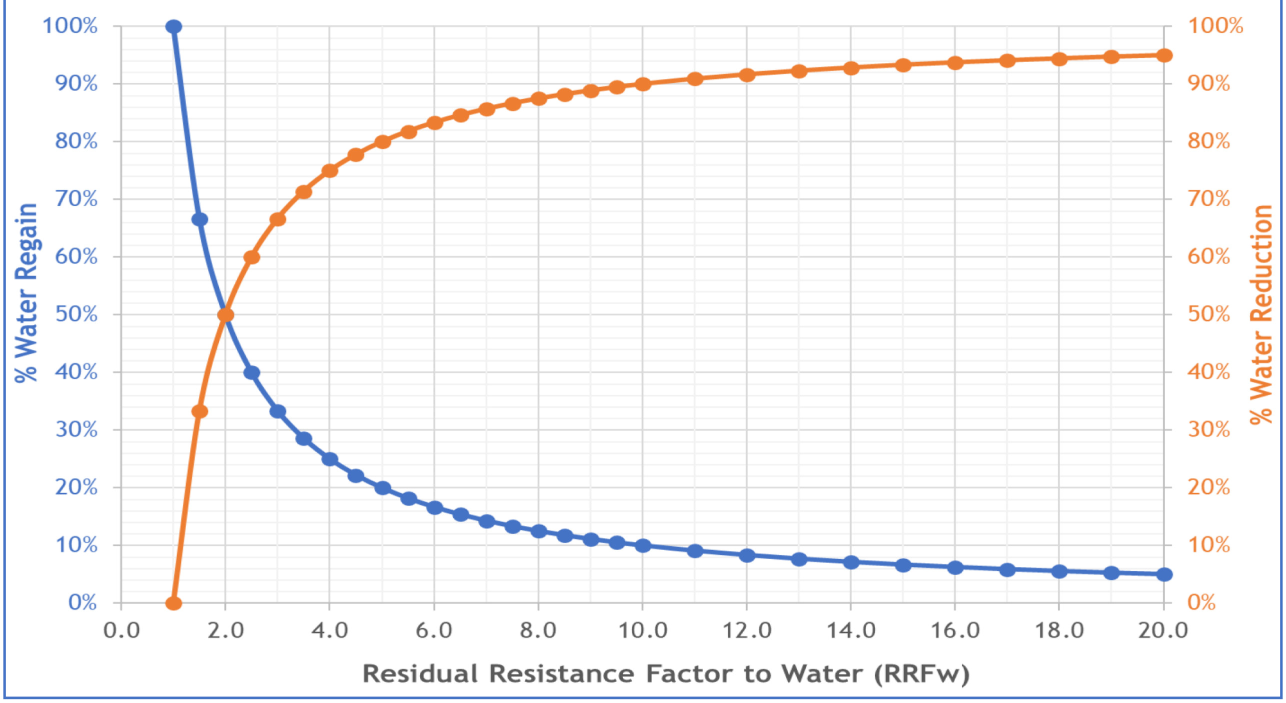


Figure 1—Relationship between RRFw and % Water Reduction

RRF_o	1.0	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2.0
% Oil Regain	100.0%	90.9%	83.3%	76.9%	71.4%	66.7%	62.5%	58.8%	55.6%	52.6%	50.0%
% Oil Reduction	0.0%	9.1%	16.7%	23.1%	28.6%	33.3%	37.5%	41.2%	44.4%	47.4%	50.0%

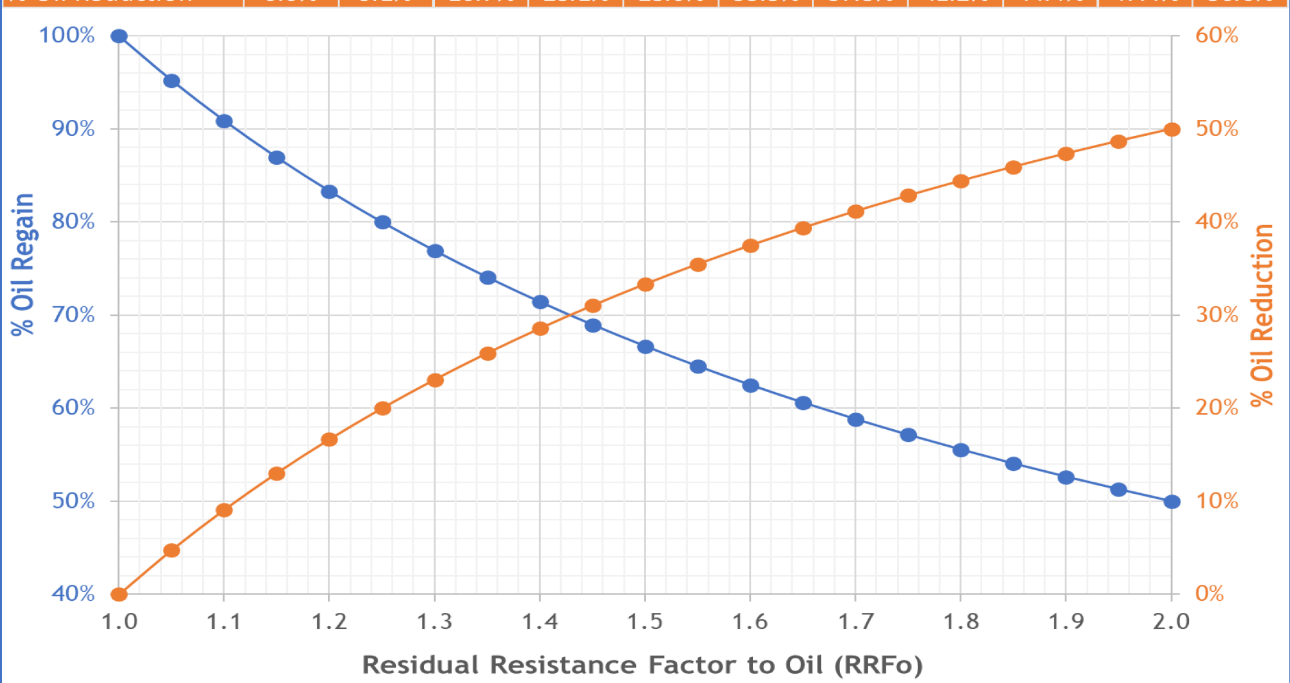


Figure 2—Relationship between RRFo and % Oil Reduction

Next Generation RPM

In the presence of oil, the RPM components deform and minimize the restriction of the formation pore throat, allowing oil or gas to flow unimpeded. Alternatively, in the presence of water, the water-wetting polymer expands, filling the pore throats which increases the resistivity of water flow. RPM can be bullheaded or pumped through coiled tubing (CT) to target and dramatically decrease the water phase of produced fluids, including high-salinity brines. Although best used for reservoirs producing excessive water through matrix flow, RPMs can also be used for temporary mitigation of water production through water coning (1).

The ability to selectively restrict water, versus shutting off all fluid flow, helps reduce costs while extending the economic life of the well. Wells that may have been shut in due to surface water handling/ disposal can now be revitalized.

Oil wells experimental procedure

Core plugs selection

The laboratory tests were performed following standardized industry testing protocols and the core plug selected with properties as per Table 1 below:

Table 1—Oil Well Core Plug

Plug #	Permeability range, md	Porosity range, %
139	49 to 69	12 to 15

Oil Compatibility/Non-Emulsion Procedure

The purpose of the test is to ensure compatibility/emulsion tendency of the formation crude oil with the aqueous-based treatment fluids, under simulated formation conditions.

The Oil Compatibility/Non-Emulsion Procedure aims to evaluate the compatibility and emulsion tendency of formation crude oil with aqueous-based treatment fluids under simulated formation conditions. The test procedure is conducted as follows: Initially, hydrocarbon-based fluids, specifically crude oil samples, are placed in a pressurized cell and preheated in an oven to a designated test temperature of 190°F. Concurrently, an equal volume of aqueous-based fluids is prepared and similarly heated to 190°F. Three distinct samples of oil-based and aqueous-based mixtures are then prepared in ratios of 75/25, 50/50, and 25/75, utilizing either a graduated cylinder or test tubes for smaller volumes. Each mixture is thoroughly agitated by hand for one minute, after which they are placed in a preheated water bath or oven within the pressurized cell, and a stopwatch is initiated. After a duration of 30 minutes, the volume of aqueous fluid that separates out (denoted as Vx) is measured at the bottom of the cylinder. Observations are recorded regarding the appearance of the aqueous/oil interface, the condition of the aqueous fluid (e.g., cloudy or clear), and any oil adhering to the walls of the cylinder. Finally, the percentage breakout is calculated using a specified equation.

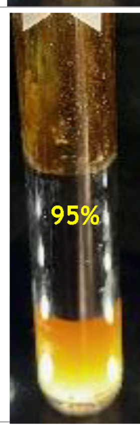
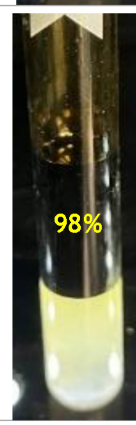
$$\%\, Breakout = \left[\frac{\mathrm{V}_x}{\mathrm{V}_0 + \mathrm{V}_f}\right] \times 200$$

Where V_f = Initial filtrate volume, ml
Vo = Initial oil volume, ml
Vx = Measured filtrate volume, ml

Oil Compatibility/Non-Emulsion Testing Results

The table 2 below show the compatibility test results for the Pre-flush with crude oil at 190°F, at different mix ratios.

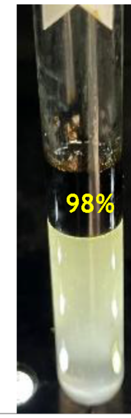
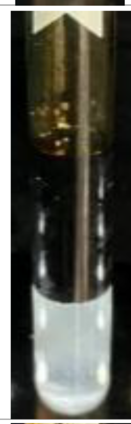
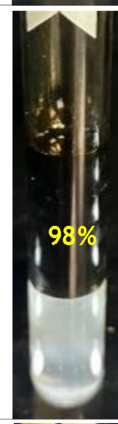
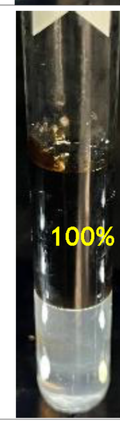
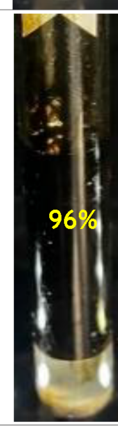
Table 2—Pre-flush / crude oil @ 190°F

Mix Ratio	0 min	5 min	10 min	30 min	60 min	180 min
75/25						
50/50						
25/75						

After 30 minutes, there was >90% separation of the aqueous filtrate phase with clean interface indicating acceptable compatibility.

The table 3 below shows the compatibility test results for the RPM with crude oil at 190°F, at different mix ratios.

Table 3—RPM / crude oil @ 190°F

Mix Ratio	0 min	5 min	10 min	30 min	60 min	180 min
75/25						
50/50						
25/75						

After 30 minutes, there was >90% separation of the aqueous filtrate phase with clean interface indicating acceptable compatibility.

Oil Wells Scenario: Core flooding

The objective of core flood testing is to assess the impact of RPM polymer system in reducing the permeability to water compared to the reduction of hydrocarbon flow.

The objective of core flooding testing is to evaluate the effectiveness of the RPM polymer system in reducing water permeability relative to its impact on hydrocarbon flow. The procedure commences with vacuum saturation of a reservoir core plug in synthetic formation brine or formation water for a minimum duration of 30 minutes, followed by loading the core plug into a core holder. A reserv oir core plug, possessing an appropriate permeability range, is inserted into the pre-cleaned flow system, where a confining pressure of 1500 psi is applied. Subsequently, back pressure of 200 psi is introduced, and any residual gas in the system is purged by flowing brine through the top and bottom flow paths. The system is then heated to the desired temperature of 190°F while maintaining back pressure, allowing both temperature and pressure to stabilize for a minimum of four hours prior to test initiation. In the production direction, specific brine permeability (Kw), using 100% formation synthetic brine, is measured at a minimum of three different flow rates for at least ten pore volumes, while recording pressure versus time data to ensure stable differential pressure with no more than 5% variation. Following this, a minimum of 50 to 100 pore volumes of synthetic mineral oil is injected at three rates until the effective oil permeability (Ko) stabilizes. Additionally, a minimum of 20 pore volumes of formation brine is injected at a fixed rate until the effective brine permeability is established. After core removal, the sample undergoes centrifugation at 200 psi capillary pressure for one hour to create an irreducible oil saturation condition. The flow system is then allowed to cool, and fluid flow is stopped before releasing back pressure and closing all valves. The sample is quickly unloaded and placed under brine, followed by reloading into a centrifuge where it is spun at 1500-2000 rpm to achieve sufficient capillary pressure for one hour. The sample is flipped in the centrifuge cup and spun again to evenly redistribute fluids, after which it is unloaded and vacuum saturated before being reintroduced into the flow system. The overburden pressure of 1500 psi is re-established, and back pressure of 200 psi is re-applied while purging residual gas and reheating the flow system.

In the production direction, 20 to 50 pore volumes of formation brine are injected at three rates until the brine permeability at irreducible oil saturation stabilizes, serving as a baseline for calculating the recovery of brine permeability. If necessary, the flow volume is extended to maintain stable differential pressure with no more than 5% variation over ten pore volumes. In the injection direction, 10 pore volumes of treatment pre-flush are injected at a fixed rate, followed by the injection of 10 pore volumes of the RPM treatment at a fixed rate. Following these injections, the bypass is opened, fluid flow is halted, data acquisition is stopped, and all valves for the sample are locked. The sample is allowed to sit for 16 to 24 hours under the specified temperature and pressure conditions. Subsequently, in the production direction, a minimum of 20 to 50 pore volumes of formation brine is injected at three rates until the post-treatment brine permeability at irreducible oil saturation remains stable, with flow volume extensions as necessary to achieve stable differential pressure. This process is followed by the injection of a minimum of 50 to 100 pore volumes of synthetic mineral oil at three rates until the post-treatment oil permeability at irreducible water saturation stabilizes, again extending the flow volume if required to maintain stable differential pressure. Steps regarding brine and oil permeability measurements are repeated in a second cycle for both phases. Finally, for the third cycle, only the brine measurement is performed. The permeability to brine is calculated using Darcy's law for linear flow, expressed as:

$$k = \frac{245 \times \mu \times \mathrm{L} \times \mathrm{Q}}{\Delta\mathrm{P} \times \mathrm{A}}$$

Where k is permeability (md), μ is the fluid viscosity (cp), L is the core plug length (cm), Q is the fluid flow rate (cc/min), ΔP is the differential pressure (psi) across the core plug and A is the core plug cross sectional area (cm^2).

Moreover, by using Darcy's law to compare the initial and final conditions of the core plug, the regain permeability (%) can be calculated as follows:

$$Regain\, k = \left[\frac{\Delta\mathrm{P}_{initial}}{\Delta\mathrm{P}_{final}}\right] \times 100$$

General Core flooding methodology

The differential pressures for the sequential flow of oil and water phases at constant rate, before and after the RPM treatment, are translated into regain permeability and Residual Resistance Factors (RRF) to water and oil. These are the parameters that will used to measure the success of the RPM in controlling water production while maintaining the flow of oil.

Additionally, the tests include multiple cycles of water and oil, after the RPM treatment to demonstrate treatment longevity since some RPM's can gradually be removed resulting in higher brine permeability over time with extended flow.

Oil Well Scenario: Core flooding results

Figure 3, shows the coreflooding results for the RPM treatment in core plug#139. Comparing the differential pressure measurements of the synthetic brine at irreducible oil saturation @ 1 cc/min before treatment (14.0 psi) and after treatment (145.2 psi), the calculated regain permeability to water is 9.6% (or a water reduction of 90.4%).

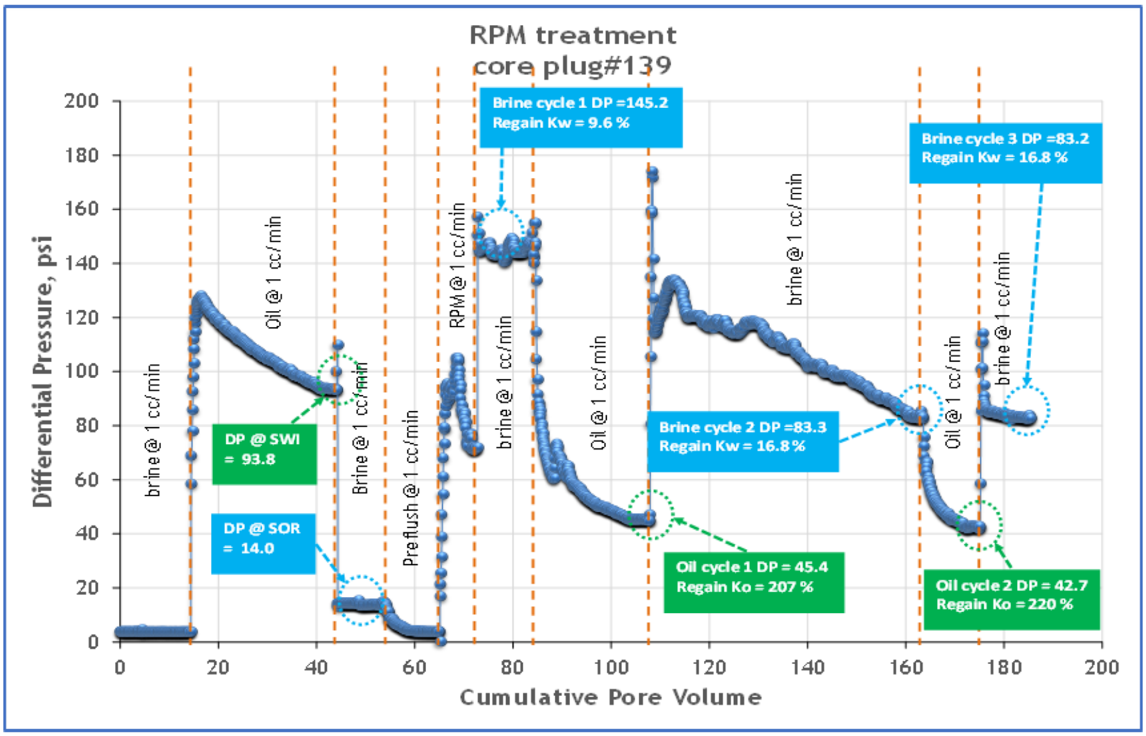


Figure 3—Core flooding results in core plug #129

Similarly, given the differential pressure measurements of the crude oil at irreducible water saturation @ 1 cc/min before treatment (93.8 psi) and after treatment (45.4 psi), the calculated regain permeability to oil is 207% (or an improvement in oil flow). This latter result is not expected and most likely as a result of the initial oil injection between 14 to 44 cummulative pore volumes not fully reaching stable conditions.

The brine and oil was reinjected (cycle 2) sequentially again to confirm the treatment performance and assess its longevity with extended fluid flow. The calculated regain permeability to water for cycle 2 is 16.8% (water reduction of 83.2%), with a regain permeability to oil of 220%. A third cycle brine regain permeability remained unchanged, 16.8%.

The following table 4 provides a summary of the core flood results for the endpoints at the various fluid stages.

Table 4—Core flooding results summary

Fluid Stage	Flowrate (cc/min)	Differential pressure (psi)	Regained Permeability (%)
Brine	1.0	3.8	-
Oil	1.0	93.8	-
Brine	1.0	14.0	-
Preflush	1.0	NA	-
RPM	1.0	105	-
Brine (Cycle 1)	1.0	145.2	9.6%
Oil (Cycle 1)	1.0	45.4	207%
Brine (Cycle 2)	1.0	83.3	16.8%
Oil (Cycle 2)	1.0	42.7	220%
Brine (Cycle 3)	1.0	83.2	16.8%

The summary of results presented in the table 5 for core plug#139 shows a very good performance, although there seems to be a 40% reduction in the water control performance with extended flow.

Table 5—RPM Performance in Oil Well core plug

RPM Performance Criteria	Parameter	Measured Values
$RRF_w \geq 5$	$RRF_{w\_Cycle\ 1}$	10.4
	$RRF_{w\_Cycle\ 2}$	6.0
	$RRF_{w\_Cycle\ 3}$	5.9
$RRF_o < 2$	$RRF_{o\_Cycle\ 1}$	0.5
	$RRF_{o\_Cycle\ 2}$	0.5
$NFRR \geq 2.5$	$NFRR_{\_Cycle\ 1}$	20.8
	$NFRR_{\_Cycle\ 2}$	12.0
Optimum RPM ≥ 0.3	Optimum $RPM_{\_Cycle\ 1}$	1.9
	Optimum $RPM_{\_Cycle\ 2}$	1.8

Fluids placement simulation

The practice of fluids placement simulation has evolved significantly since its inception, simulations play a crucial role in optimizing treatment design, predicting fluid flow, and minimizing costs.

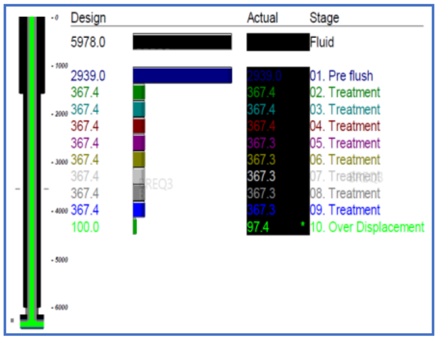
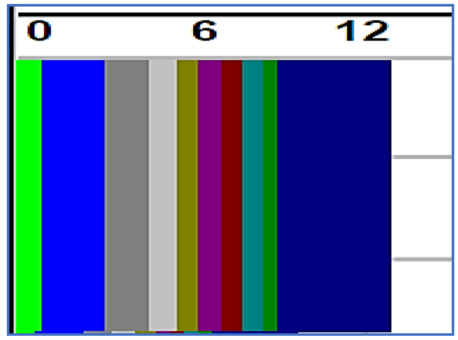
The model accounts for the pressure and flow rate of the acid injection, which influence the depth and uniformity of non-reactive fluids penetration, incorporates properties of the fluids, such as its concentration, viscosity, and kinetics, to accurately model it and it may also account for variations in rock properties, such as permeability, porosity, and mineral composition, which affect penetration and reaction rates.

The equation that the model utilizes for the radial penetration calculations is the following:

Treatment volume (gal) = 7.4805 * porosity * Desired treatment radius (ft) – wellbore radius (ft) * Height (ft) * 3.1416

A simulation showing the fluids placement simulation of an oil well is presented in the following table 6.

Table 6—Final Fluid placement simulation

Fluids injection pattern	Final fluids placement (ft)
	

The figure 4 and table 7 presents the sensitivity analysis done considering several scenarios of water rate reduction on a hypothetical case of an Oil well.

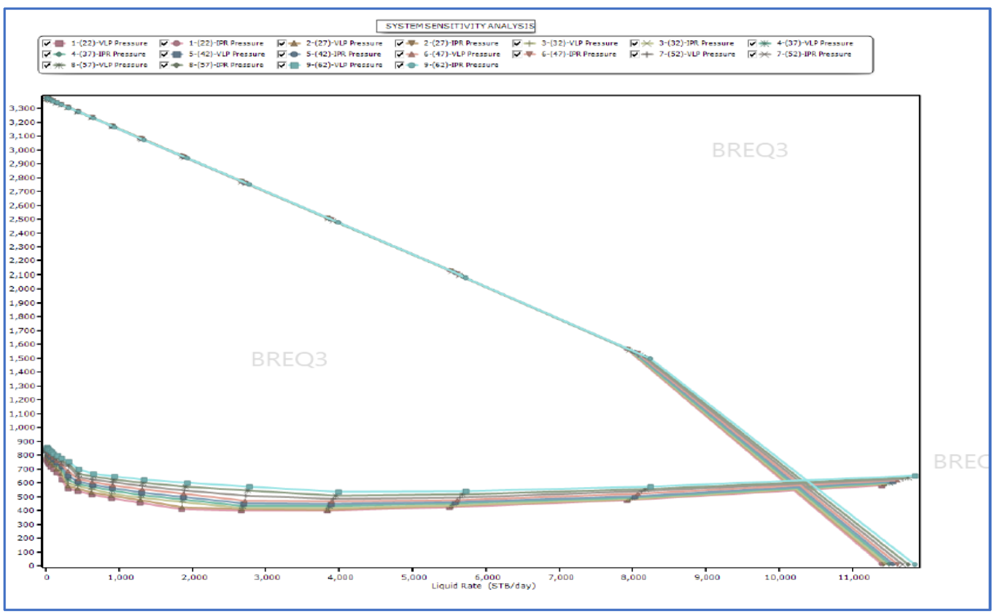


Figure 4—Oil & Water production analysis

Table 7—Water rate reduction (%)

Water Rate Reduction (%)
-
7.9
15.9
23.8
31.8
39.8
47.8
55.9
63.9

Gas Well Scenario: Core flooding results

Core plugs were selected with petrophysical properties shown below. For preparation, the plugs were cleaned using Soxhlet equipment and methanol then taken out of cleaning after no presence of Chloride was observed on the liquid at the extraction chamber while tested with Silver Nitrate.

Table 8—Gas Well core plugs properties

Plug#	Permeability range, md	Porosity range, %
43	4.1 to 11	7.1 to 9

Core flooding test for Gas Well Scenario

The objective of core flood testing is to assess the impact of RPM polymer system in reducing the permeability to water compared to the reduction of hydrocarbon flow.

The core sample was saturated with 5 wt% KCl brine in a vacuum and pressure saturator and then loaded into the core holder. Note: The 5 wt% KCl brine was filtered with a 2.5 µm pore size before testing. Confining pressure of 2000 psi, pore pressure of 500 psi, and a temperature of 275 °F were applied during the core flooding test. Permeability to 5 wt% KCl brine in the production direction (from formation to wellbore direction) was measured until constant values were obtained at 0.50 cc/min. Note: For all permeability measurements, more than 5 pore volumes were injected, showing stable readings. This measurement was repeated at 0.75 cc/min to ensure consistency.

Subsequently, permeability to humidified nitrogen gas in the production direction was measured until constant values were obtained at around 6 cc/min, and this was repeated at around 8 cc/min. The permeability to 5 wt% KCl brine in the production direction was measured again at 0.50 cc/min and 0.75 cc/min. For the evaluation of RPM, a total of 10 pore volumes were injected in the wellbore to formation direction, and the differential pressure versus pore volume values were recorded. After injecting RPM, the core plug was left in static conditions to soak at temperature for at least 12 hours.

Following this, the permeability to 5 wt% KCl brine in the production direction was measured again at 0.50 cc/min and 0.75 cc/min. Permeability to humidified nitrogen gas in the production direction was also measured again at around 6 cc/min and 8 cc/min. Finally, permeability to 5 wt% KCl brine in the production direction was measured once more at 0.50 cc/min and 0.75 cc/min. Optionally, an acid treatment could be applied in the injection direction (wellbore to formation direction) at 0.75 cc/min, and the permeability to 5 wt% KCl brine in the production direction was measured again at 0.75 cc/min.

Finally, the system temperature was taken down to room temperature while keeping the sample under pore pressure conditions, and the sample was offloaded after releasing pore pressure. The permeability to brine was calculated using Darcy's law for linear flow:

$$k = \frac{245 \times \mu \times L \times Q}{\Delta P \times A}$$

Where k is permeability (md), µ is the fluid viscosity (cp), L is the core plug length (cm), Q is the fluid flow rate (cc/min), ΔP is the differential pressure (psi) across the core plug and A is the core plug cross sectional area (cm^2). Using Darcy's law to compare the initial and final conditions of the core plug, the regain permeability (%) is calculated as follows:

$$Regain\ k = \left[\frac{\Delta P_{initial}}{\Delta P_{final}}\right] \times 100$$

Figure 5, shows the coreflooding results for the RPM treatment in core plug#43. Comparing the differential pressure measurements of the KCl brine at irreducible gas saturation at 0.5 and 0.75 cc/min before treatment (4.8 and 7.2 psi) and after treatment (64.9 and 96.0 psi), the calculated average regain permeability to water is 7.5% (or a water reduction of 92.5%).

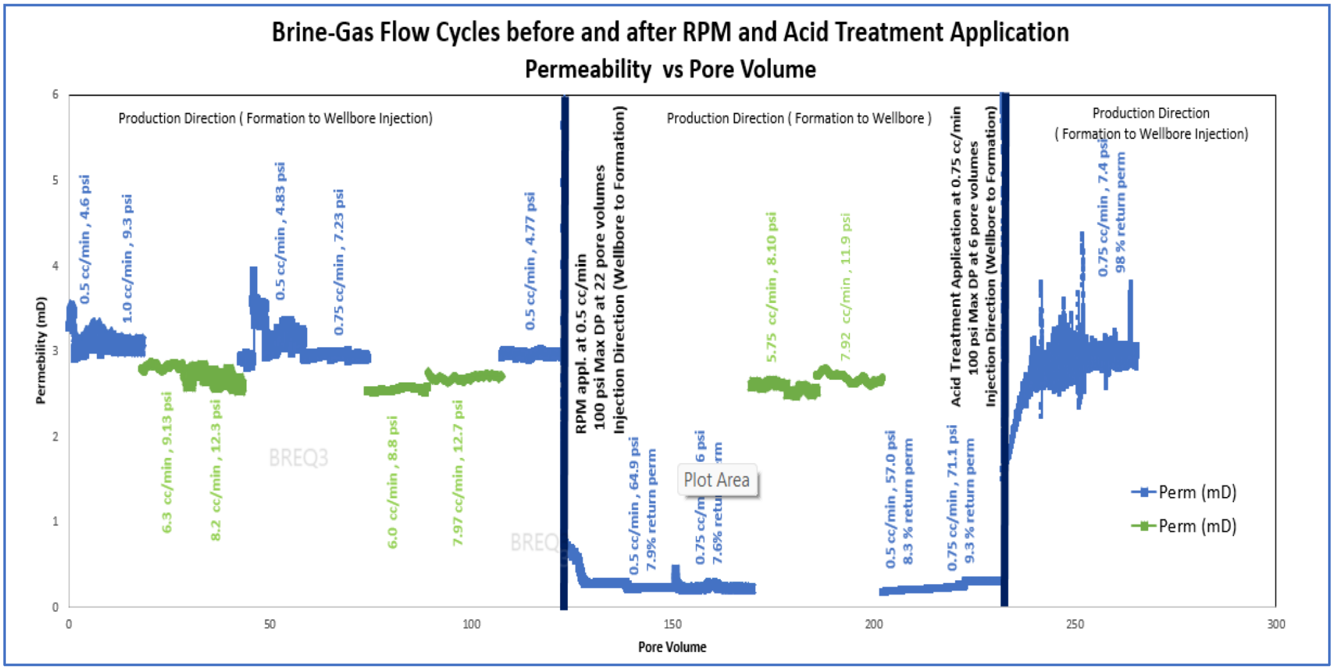


Figure 5—Core flooding for plug #43

Similarly, given the differential pressure measurements of the N2 gas flow at irreducible water saturation at ~ 6 and 8 cc/min before treatment (8.8 and 12.7 psi) and after treatment (8.1 and 11.9 psi), the calculated average regain permeability to gas is 106.7% (or a slight improvement in gas flow). The brine was reinjected (cycle 2) to confirm the treatment performance and assess its longevity with extended fluid flow. The calculated regain permeability to water for cycle 2 is 9.3% (water reduction of 90.7%). A third cycle brine regain permeability after an acid treatment confirms that its capable of restoring any polymer damage, with regain permeability to water of 97.3%.

The following table 9 provides a summary of the core flood results for the endpoints at the various fluid stages.

Table 9—Core flooding results summary

Fluid Stage	Flowrate (cc/min)	Differential Pressure (psi)	Regained Permeability (%)
5 wt% KCl brine	0.5	4.6	---
5 wt% KCl brine	1.0	9.3	---
N_2 gas	6.3	9.1	
N_2 gas	8.2	12.3	
5 wt% KCl brine	0.5	4.8	
5 wt% KCl brine	0.75	7.2	
N_2 gas	6.0	8.8	
N_2 gas	8.0	12.7	
5 wt% KCl brine	0.5	4.8	
0.5 vol% RPM	0.5	100 (max)	
5 wt% KCl Brine (Cycle 1)	0.5	64.9	7.4%
5 wt% KCl Brine (Cycle 1)	0.75	96.0	7.5%
N_2 gas (Cycle 1)	5.8	8.1	108.6%
N_2 gas (Cycle 1)	7.9	11.9	106.7%
5 wt% KCl Brine (Cycle 2)	0.5	57.0	8.4%
5 wt% KCl Brine (Cycle 2)	0.75	71.1	10.1%
Acid Treatment	0.5	23 (max)	NA
5 wt% KCl Brine (Cycle 3)	0.75	7.4	97.3%

The summary of results presented in the table 10 for core plug#43 shows a very good performance.

Table 10—RPM Performance in Gas Well core plug

RPM Performance Criteria	Parameter	Measured Values
$RRF_w \geq 5$	$RRF_{w\_Cycle\ 1}$	13.4
	$RRF_{w\_Cycle\ 2}$	10.9
	$RRF_{w\_Cycle\ 3\ (acid)}$	1.0
$RRF_g < 2$	$RRF_{g\_Cycle\ 1}$	0.9
$NFRR \geq 2.5$	$NFRR_{\_Cycle\ 1}$	14.9
	$NFRR_{\_Cycle\ 2}$	NA
Optimum RPM ≥ 0.3	Optimum RPM $_{\_Cycle\ 1}$	1.0
	Optimum RPM $_{\_Cycle\ 2}$	NA

Fluids placement simulation for Gas well scenario

The model accounts for the pressure and flow rate of the acid injection, which influence the depth and uniformity of non-reactive fluids penetration, incorporates properties of the fluids, such as its concentration, viscosity, and kinetics, to accurately model it and it may also account for variations in rock properties, such as permeability, porosity, and mineral composition, which affect penetration and reaction rates.

The equation that the model utilizes for the radial penetration calculations is the following:

Treatment volume (gal) = 7.4805 * porosity * Desired treatment radius (ft) – wellbore radius (ft) * Height (ft) * 3.1416

A simulation showing the fluids placement simulation of a gas well is presented in the figure 6.

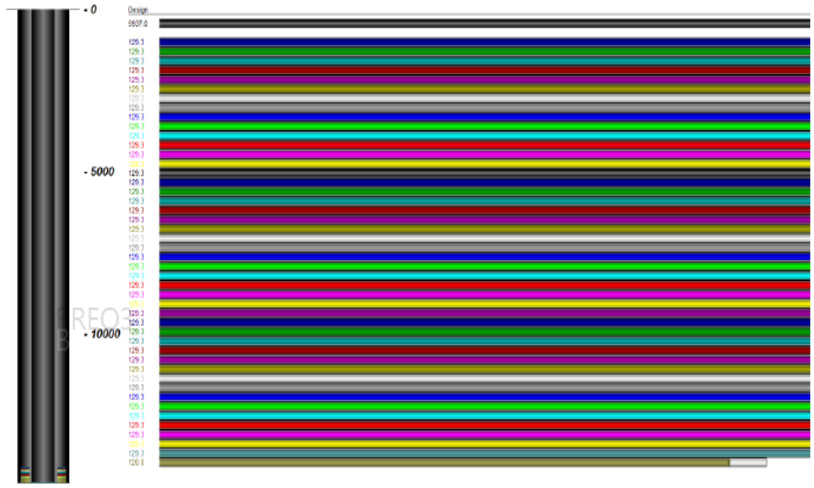


Figure 6—Final Fluid placement simulation

The figure 6 and table 11 presents the sensitivity analysis done considering several scenarios of water rate reduction on a hypothetical case of a Gas well.

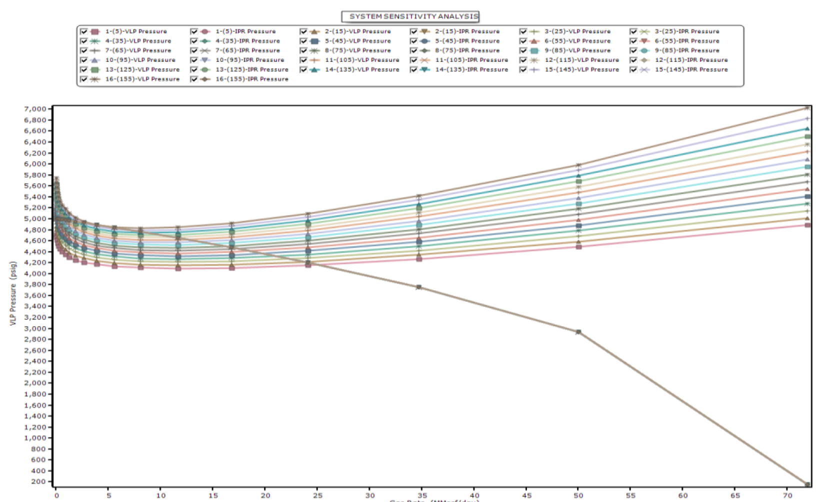


Figure 7—Gas & Water production analysis

Table 11—Water rate reduction (%)

WGR (STB/MMSCF)	Water Rate Reduction (%)
155	-
145	3.7
135	8.0
125	12.7
115	17.7
105	23.0
95	28.7
85	34.7
75	41.1
65	47.8
55	54.9
45	62.3
35	70.1
25	78.2
15	86.6
5	95.4

Future Trends

Increased adoption: As the technology matures and costs decrease, the adoption of RPM's is expected to increase across the oil and gas industry.

Advanced sensors: The development of advanced sensors integrated with fiber optics will further enhance the capabilities to effectively trace the RPM's injection in real time.

AI and machine learning: The integration of artificial intelligence and machine learning with fiber optic data will enable predictive analytics and automated decision-making during water control treatments

Conclusions

- RPM treatment and pre-flush fluid systems are compatible with the crude oil sample resulting in no emulsion.
- RPM showed good water control performance in carbonate core plugs up to 190 deg F
- The Residual Resistance Factor to water (RRFw) varied from a high of 10.4 to a low of 2.6 versus the Residual Resistance Factor to oil (RRFo) between 2.4 to 0.5
- Longevity of RPM treatments can potentially be affected by very high production rates with superficial fluid velocity > 0.345 in/min and/or produced waters with pH to ≤ 5
- In core plug #43, the permeability to water was reduced by 92.5 and 90.7%, after successive cycles of 5 wt% KCl brine whilst the regain permeability to N2 gas remain unaffected with a value of 107.7%
- The core flooding test performed on the core #43 demonstrate that the RPM performance criteria for the test performed in the core #43 meets the RRFw = 10.9 to 13.4 (target ≥ 5) and RRFg = 0.9 (target < 2).
- Acid proved successful in removing the effects of the RPM in the core plug #43, with a brine regained permeability of 97.3%

Some important considerations for good candidates for RPM treatments are as follows:

1. No reservoir crossflow
2. Excessive water production flow from zones with segregated oil and water flow is preferred, but other mechanisms may also be considered
3. Water production flow is through matrix permeability, not from natural fractures or fissures
4. Production water pH should be greater than 5
5. If possible, select candidates with formation water TDS below 200k mg/L

References

1. Patent number: 6476169 Methods of reducing subterranean formation water permeability Larry, S. Eoff, B. Raghava Reddy, Eldon D. Dalrymple
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